

Test of Mathematics for University Admission

Original Mock Examination 01

Student edition Paper 1

Time: 75 minutes

Additional materials: Answer sheet

Instructions to Candidates

Read these instructions carefully. Do not turn the page until you are told to begin.

A separate answer sheet is provided. Check that you have one. You will also need a soft pencil and an eraser.

This is the first of two papers.

There are 20 questions. For each question, choose one answer and record it on the answer sheet. If you change an answer, erase your previous mark thoroughly.

Each question is worth one mark. There are no penalties for incorrect answers, so you should attempt all 20 questions.

Do all rough work on this question paper. No extra paper is allowed.

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1 What is the product of all real solutions of

$$|x - 2| + |x + 2| = x + 6?$$

A -4

B -12

C 4

D 12

E 24

2 An arithmetic progression has an odd number N of terms and a non-zero sum. The sum of the terms in odd-numbered positions is $\frac{5}{9}$ of the sum of all N terms. What is the value of N ?

- A 9
- B 7
- C 5
- D 11
- E 13

3 Two parallel chords of a circle have lengths 6 and 8. The perpendicular distance between the lines containing the chords is 7. What is the radius of the circle?

A 4

B $\sqrt{21}$

C 5

D $\sqrt{29}$

E 7

4 Positive integers x and y satisfy

$$31x + 14y = 1000.$$

What is the least possible value of $x + y$?

A 32

B 33

C 34

D 35

E 36

5 The equation $\sin x = k$ has two distinct solutions u and v in the interval $0 < x < \pi$. Given that

$$u^2 + v^2 = \frac{5\pi^2}{8},$$

what is the value of k ?

A $\frac{\sqrt{2}}{4}$

B $\frac{1}{2}$

C 1

D $\frac{\sqrt{3}}{2}$

E $\frac{\sqrt{2}}{2}$

6 For a real number k , the equation

$$\log_2(\log_2 x) + \log_2 \left(\log_2 \frac{16}{x} \right) = k$$

has two real solutions $x_1 < x_2$ satisfying $x_2 = 4x_1$. What is k ?

A 0

B 1

C $\log_2 3$

D 2

E 3

7 A fair coin is tossed six times. The tosses are independent.

What is the probability that, in exactly two of the five pairs of consecutive tosses, the two outcomes are the same?

A $\frac{5}{16}$

B $\frac{15}{64}$

C $\frac{5}{32}$

D $\frac{1}{2}$

E $\frac{5}{8}$

8 A monic quartic polynomial $p(x)$ satisfies

$$p(x) = x^4 p\left(\frac{1}{x}\right)$$

for every non-zero x . Given that $p(2) = p(-3) = 0$, find $p(1)$.

A $-\frac{10}{3}$

B $-\frac{8}{3}$

C -2

D $\frac{8}{3}$

E $\frac{10}{3}$

9 Let $\lfloor t \rfloor$ denote the greatest integer less than or equal to t . The sequence (a_n) is defined by

$$a_n = \lfloor n\sqrt{2} \rfloor - \lfloor (n-1)\sqrt{2} \rfloor.$$

How many of $a_1, a_2, \dots, a_{100}$ are equal to 2?

A 39

B 40

C 43

D 42

E 41

10 The function f is defined by $f(x) = x^3 + x$ for all real x , and g is its inverse. A real number t satisfies

$$g(t) + g(8 - t) = 2.$$

What is the value of $g(t)g(8 - t)$?

A $-\frac{1}{3}$

B 0

C $\frac{2}{3}$

D $\frac{1}{3}$

E 1

11 A monic polynomial $f(x)$ of degree 4 satisfies

$$f(0) = 0, \quad f(1) = 1, \quad f(2) = 4, \quad f(3) = 9.$$

Find $f(4)$.

A 16

B 40

C 32

D 24

E 48

12 The real numbers a and b are such that the coefficient of x in the expansion of

$$(1 + ax)^2(1 + bx)$$

is 3.

What is the greatest possible value of the coefficient of x^2 in the expansion of

$$(1 + ax)^4(1 + bx)?$$

A 9

B 12

C 18

D 24

E There is no greatest value.

13 An angle θ is chosen uniformly at random from

$$\{0^\circ, 15^\circ, 30^\circ, \dots, 345^\circ\}.$$

Given that $\sin \theta > \frac{1}{2}$, what is the probability that $\cos \theta < \frac{1}{2}$?

A $\frac{5}{24}$

B $\frac{2}{3}$

C $\frac{3}{4}$

D $\frac{5}{7}$

E $\frac{6}{7}$

14 How many ordered pairs of integers (x, y) satisfy both of the following inequalities?

$$x^2 + 2xy + y^2 - 10x - 10y + 9 \leq 0,$$

$$x^2 - 2xy + y^2 - 10x + 10y + 16 \leq 0.$$

A 31

B 28

C 32

D 56

E 63

15 A point P on the curve $y = x^3$ has positive x -coordinate. The tangent at P meets the curve again at Q , and the tangent at Q meets the curve again at R . The x -coordinates of P and R differ by 3. What is the gradient of the tangent at P ?

A $\frac{1}{3}$

B 1

C 27

D 9

E 3

16 A function f is integrable on every finite interval and satisfies $f(x+1) = f(x) + 1$ for all real x . Given that

$$\int_0^1 f(x) dx = 1, \quad \int_0^1 f(x)^2 dx = 2,$$

what is $\int_0^3 f(x)^2 dx$?

A 6

B 17

C 12

D 15

E 9

17 Let $f(x) = x + \sin x$. The real numbers $a_1, a_2, \dots, a_7$ form an arithmetic progression with common difference $\pi/3$. Given that

$$\sum_{k=1}^7 f(a_k) = 7\pi,$$

what is the value of

$$[f(a_4)]^2 - a_2 a_6?$$

A $\frac{4\pi^2}{9}$

B $\frac{\pi^2}{9}$

C $\frac{2\pi^2}{9}$

D 0

E $\frac{5\pi^2}{9}$

18 The graph of a cubic polynomial f has two distinct stationary points with x -coordinates u and v . The line joining these two points has gradient -4 . What is the gradient of the tangent to the graph at $x = \frac{u+v}{2}$?

A -8

B 0

C -4

D -2

E -6

19 The function f is defined by

$$f(x) = \frac{x^3 - c^3}{x - c} \quad (x \geq 0),$$

where $c < 0$. The points A , B and C on the graph have x -coordinates 0, a and 3, respectively, where $0 < a < 3$. The chord AB is horizontal. The area enclosed by AB and the curve is eight times the area enclosed by the chord BC and the curve. What is c ?

A $-\frac{1}{2}$

B -1

C $-\frac{3}{2}$

D -2

E $-\frac{5}{2}$

20 In triangle ABC , the internal angle bisector from A meets the side BC at E . The ray AE meets the circumcircle of ABC again at D , beyond E . If $AE = 1$, $ED = 8$ and $BC = 6$, what is the area of triangle ABC ?

A $\frac{3\sqrt{7}}{8}$

B $\frac{3\sqrt{15}}{4}$

C $\frac{9\sqrt{7}}{8}$

D 3

E $\frac{9\sqrt{7}}{4}$

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1 Consider the following statement about a polynomial p :

If $p(n) \geq 0$ for every integer n , then $p(x) \geq 0$ for every real x .

Which of the following polynomials is a counterexample to this statement?

A $p(x) = x^2 + 1$

B $p(x) = x^2 - x$

C $p(x) = x^2 - x + 1$

D $p(x) = (x - 1)^2$

E $p(x) = x^4 + x^2$

2 Let a , b and c be positive real numbers, none of which is equal to 1, such that

$$\log_a b + \log_b c = 0.$$

Which statement must be true?

- A The numbers a and c are both greater than 1.
- B The numbers a and c are both less than 1.
- C Exactly one of a and c is greater than 1.
- D The number b lies strictly between a and c .
- E $ac = 1$.

3 A circle passes through $(0, 0)$ and $(4, 0)$. Which of the following conditions is/are individually sufficient to determine the circle uniquely?

I. Its area is 9π .

II. Its centre lies on the line $y = x$.

III. Its radius is 2.

A None of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

4 Let a and b be fixed real constants. Consider the equation

$$a \sin^2 x + b \cos^2 x = 1.$$

Which of the following pairs of values of x has the property that, if the equation holds at both values, it must hold for every real x ?

A $x = \pi/6$ and $x = 5\pi/6$.

B $x = \pi/4$ and $x = 3\pi/4$.

C $x = \pi/3$ and $x = 2\pi/3$.

D $x = \pi/6$ and $x = \pi/3$.

E $x = \pi/6$ and $x = 7\pi/6$.

5 A batch passes a test if every measurement in it is less than 10. Every batch that passes is labelled green. A particular batch is not labelled green. Which statement must be true?

- A Every measurement in the batch is at least 10.
- B At least one measurement in the batch is greater than 10.
- C At least one measurement in the batch is at least 10.
- D At least one measurement in the batch is less than 10.
- E No measurement in the batch is equal to 10.

6 The diagonals of a convex quadrilateral $ABCD$ intersect at P . Triangles APB and CPD have equal areas, and triangles APD and BPC have equal areas. Which of the following must be true?

A $ABCD$ is a rectangle.

B $ABCD$ is a rhombus.

C $ABCD$ is a parallelogram.

D The four vertices lie on a circle.

E AC is perpendicular to BD .

7 The sequence (a_n) is defined by

$$a_n = a + (n - 1)d,$$

where a and d are positive integers. The sequence is called self-summing if, for every positive integer n , there is a positive integer b_n such that

$$ab_n = a_1 + a_2 + \cdots + a_n.$$

Consider the following statements.

- I. The sequence is self-summing if and only if $d \mid a$.
- II. If the sequence is self-summing, then $b_{n+1} - b_n = n + a/d$.
- III. If the sequence is self-summing, then (b_n) is an arithmetic progression.

Which of these statements must be true?

- A None of them
- B I only
- C II only
- D III only
- E I and II only
- F I and III only
- G II and III only
- H I, II and III

8 Let n be a positive integer which is not a perfect square, and let

$$x = \sqrt{n+1} - \sqrt{n}.$$

Consider the following statements.

- I. x is irrational.
- II. $1/x$ is irrational.
- III. x^2 is irrational.

Which of these statements must be true?

- A None of them
- B I only
- C II only
- D III only
- E I and II only
- F I and III only
- G II and III only
- H I, II and III

9 Let $p(x)$ be a polynomial of degree n with $p(0) \neq 0$. The polynomial $q(x)$ is obtained by expanding and simplifying

$$x^n p\left(\frac{1}{x}\right).$$

Consider the following statements.

- I. $q(x)$ is a polynomial of degree n .
- II. If r is a non-zero root of $p(x)$, then $1/r$ is a root of $q(x)$ with the same multiplicity.
- III. $q(x)$ is monic if and only if $p(0) = 1$.

Which of these statements must be true?

A None of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

10 The function $f : \mathbb{R} \rightarrow \mathbb{R}$ is strictly increasing, with $f(0) = 0$ and $f(1) = 2$. Which equation must have exactly one real solution?

A $f(x) + f(1 - x) = 2$

B $f(x) + f(x + 1) = 2$

C $f(x)f(x + 1) = 0$

D $f(x) = 2x$

E $f(f(x)) = 2$

11 Let $a_1 > 0$ and define

$$a_{n+1} = \frac{1}{2} \left(a_n + \frac{1}{a_n} \right).$$

Consider the following statements.

- I. $a_2 \geq 1$.
- II. The sequence $a_2, a_3, a_4, \dots$ is decreasing or constant.
- III. The sequence (a_n) converges to 1.

Which of these statements must be true?

A None of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

12 Consider the following statement about positive real numbers a and b :

If $a + b \leq 2$, then $ab \leq 1$.

Which pair (a, b) is a counterexample to both the converse and the inverse of this statement?

A $(\frac{1}{2}, \frac{1}{2})$

B $(1, 1)$

C $(2, 2)$

D $(\frac{1}{2}, 2)$

E $(\frac{1}{2}, \frac{3}{2})$

13 A monic cubic polynomial has three distinct positive real zeros. A tangent is drawn at each of the three points where its graph meets the x -axis. How many of these tangents meet the positive y -axis?

A 0

B 1

C 2

D 3

E The number cannot be determined.

14 Let $p(x)$ be a monic polynomial with non-zero coefficients. Consider the following statements.

- I. If all its roots are positive real numbers, then its coefficients alternate in sign.
- II. If its coefficients alternate in sign, then it has no negative real root.
- III. If its coefficients alternate in sign, then all its roots are positive real numbers.

Which of these statements must be true?

A None of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

15 Start with the list $L_0 = (1, 3)$. To obtain L_{n+1} from L_n , insert the arithmetic mean of each pair of adjacent entries between those entries. Let x_n be the sum of all entries of L_n , and define $y_n = x_n - 1$. Consider the following statements.

- I. The list L_4 contains 17 entries.
- II. $x_{n+1} = 2x_n - 2$ for every $n \geq 0$.
- III. The sequence (y_n) is a geometric progression.

Which of these statements must be true?

A None of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

16 The functions f and g are continuous on $[0, 1]$ and satisfy $f(x) \geq g(x)$ for every $x \in [0, 1]$. It is also given that

$$\int_0^1 f(x)^2 dx < \int_0^1 g(x)^2 dx.$$

Which statement must be true?

A $f(x) < 0$ for every $x \in [0, 1]$.

B $g(x) < 0$ for every $x \in [0, 1]$.

C There is an $x \in [0, 1]$ such that $f(x) + g(x) < 0$.

D $f(x) \leq -g(x)$ for every $x \in [0, 1]$.

E $\int_0^1 f(x) dx < \int_0^1 g(x) dx$.

17 A polynomial f of degree at most two maps $[0, 1]$ into $[0, 1]$, with $f(0) = 1$ and $f(1) = 0$. As f varies, what is the complete set of values of $c \in [0, 1]$ satisfying $f(c) = c$?

A $[\frac{1}{3}, \frac{2}{3}]$

B $[\frac{3-\sqrt{5}}{2}, \frac{1}{2}]$

C $[\frac{3-\sqrt{5}}{2}, \frac{\sqrt{5}-1}{2}]$

D $[\frac{1}{2}, \frac{\sqrt{5}-1}{2}]$

E $(\frac{3-\sqrt{5}}{2}, \frac{\sqrt{5}-1}{2})$

18 A real polynomial p has a strict local minimum at $x = 0$ and satisfies $p''(0) = 0$. It also has a stationary point at $x = 1$, with $p(1) < p(0)$. What is the smallest possible degree of p ?

A 3

B 4

C 5

D 6

E 7

19 How many continuous functions $f : [0, 2] \rightarrow [0, 1]$ satisfy both

$$\int_0^1 f(x)^2 dx = \int_0^1 f(x) dx, \quad \int_0^2 f(x) dx = 1?$$

- A None
- B Exactly one
- C Exactly two
- D Exactly four
- E Infinitely many

20 Let

$$(1 + ax)^n = c_0 + c_1x + \cdots + c_nx^n,$$

where $a > 0$ and n is an integer with $n \geq 4$.

Given that $c_2 = c_3$, which of the following must be true?

A $c_1 < c_4$.

B $c_1 = c_4$.

C $c_1 > c_4$.

D No coefficient is greater than c_2 .

E No coefficient is smaller than c_0 .

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